

Semantic Segmentation of Weeds in Soybean Fields

1. Introduction

The goal of this project was to develop a semantic segmentation pipeline for detecting weeds in soybean fields using two distinct approaches, Roboflow and Google Colab.

A total of 46 labeled field images were used, where 44 were pre-labeled and 2 were manually annotated.

Each image was segmented into three classes: background, soybean, and weed.

The objective was to evaluate both platforms' performance and investigate how data preprocessing and augmentation strategies affect model accuracy.

2. Methodology

2.1 Roboflow Implementation

A new project was created in **Roboflow 2.0** under the name *Weed Detection 1*.

The dataset was uploaded in segmentation mask format with the following configuration:

- **Dataset size:** 110 total images
 - Train: 96 (87%)
 - Validation: 9 (8%)
 - Test: 5 (5%)
- **Preprocessing:**
 - Auto-Orient (applied)
 - Resize: Stretch to 960×540
- **Augmentations (Configuration 1):**
 - Brightness: $\pm 15\%$
 - Saturation: $\pm 25\%$

After training the semantic segmentation model in Roboflow, performance was evaluated using mean Intersection over Union (mIoU).

Roboflow Model Performance:

- mIoU = **39.3%**

The relatively moderate accuracy suggests underfitting, possibly due to the small dataset size and simple augmentation strategy.

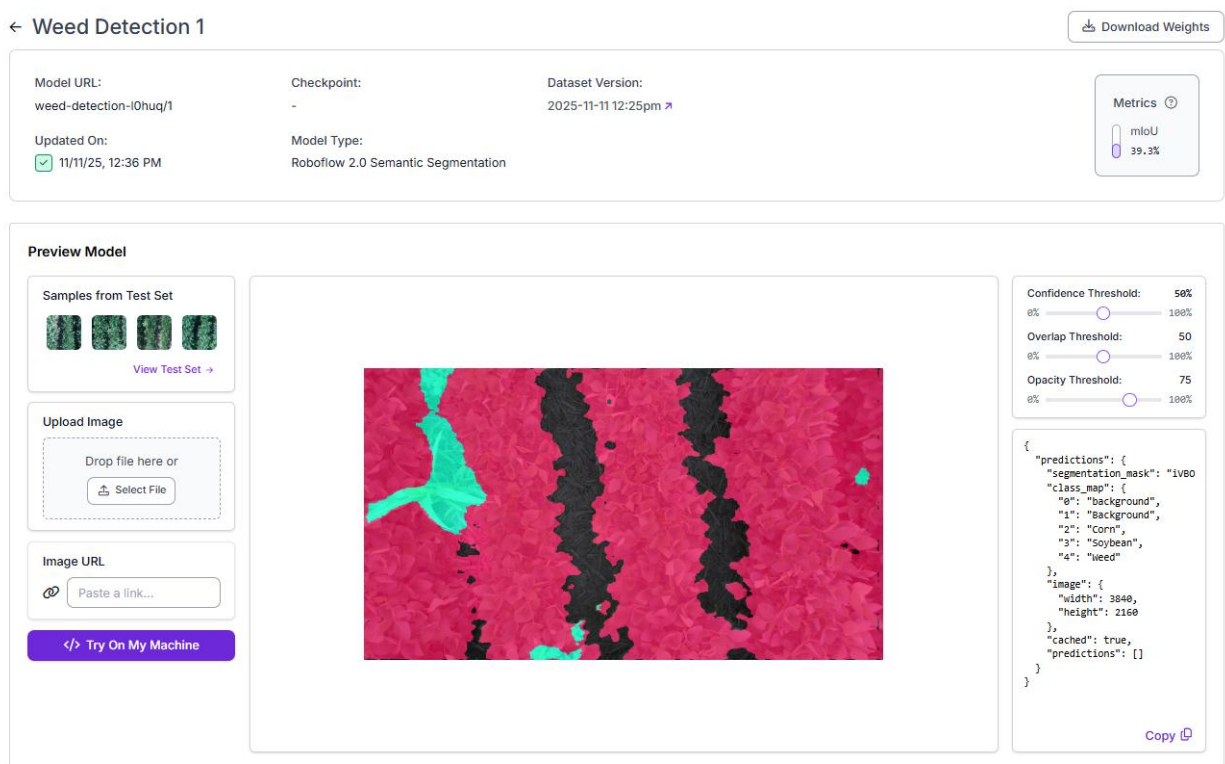


Figure 1: Roboflow Model Training.

Example visualizations show the model correctly identified major weed clusters but struggled with fine boundaries between soybean leaves and weeds.

2.2 Colab Implementation

The same dataset was exported from Roboflow in *Semantic Segmentation Mask* format and used in Google Colab. A custom data loader and visualization pipeline were implemented. The model

architecture was SegFormer-b0, pre-trained on ADE-512-512 from Hugging Face, fine-tuned for three classes (background, soybean, weed).

Training Configuration:

- Optimizer: Adam (lr = $1e-4$, weight decay = $1e-4$)
- Loss: Combined CrossEntropy + Dice Loss
- Epochs: 300 (early stopping triggered at 132)
- Batch size: 16
- Hardware: NVIDIA A100 GPU via Colab
- Metrics computed with TorchMetrics: Precision, Recall, F1, IoU

3. Results and Graphs

3.1 Training Curves (Colab)

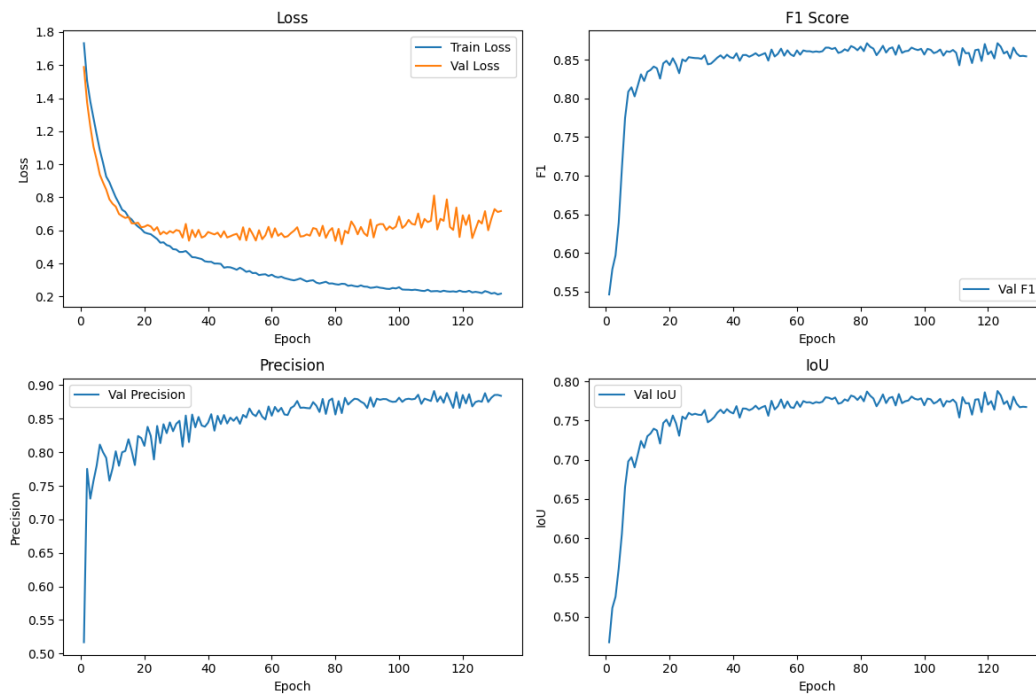


Figure 2: Training and validation loss, F1 score, precision, and IoU across epochs. Loss decreased steadily and plateaued after ~100 epochs, while validation metrics stabilized around high-0.8 ranges, indicating convergence.

3.2 Test Metrics (Colab SegFormer Model)

Metric	Score
Precision	0.8796
Recall	0.8061
F1 Score	0.8411
IoU	0.7672

3.3 Visual Results

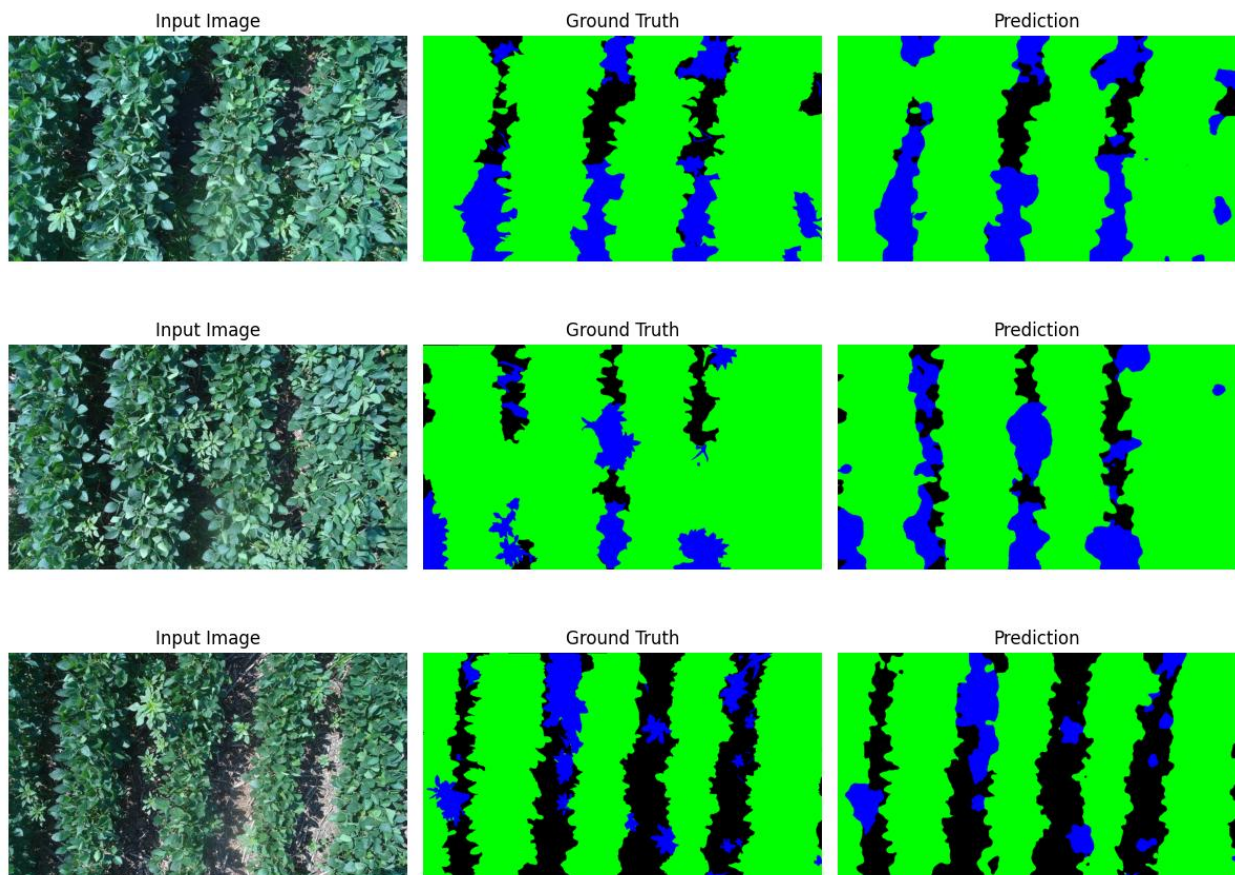


Figure 3: Example predictions from the SegFormer model showing the input image, ground truth, and predicted segmentation.

4. Comparative Discussion

Platform	Model	Augmentation	Key Metric (mIoU/F1)	Remarks
Roboflow	Roboflow 2.0 Semantic Segmentation	Brightness $\pm 15\%$, Saturation $\pm 25\%$	mIoU = 39.3%	Easier setup, but limited model capacity and augmentation diversity.
Colab (PyTorch)	SegFormer-b0 Fine-Tuned	Custom normalization, random flips, Dice + CE Loss	IoU = 76.7%, F1 = 84.1%	Significantly higher accuracy and better generalization.

Analysis:

- The Colab SegFormer model clearly outperformed Roboflow's built-in training, achieving roughly double the mIoU.
- Advanced loss functions (Dice + CrossEntropy) and GPU fine-tuning improved edge segmentation and small weed detection.
- Roboflow's preprocessing provided a convenient baseline, but stretching images to a fixed 960×540 resolution might have slightly distorted weed morphology, impacting accuracy.
- Augmentation diversity was limited in Roboflow; random flips and rotations in Colab improved robustness.
- The dataset size (46 labeled images) remained a limiting factor in both methods, constraining generalization on unseen test images.

5. Conclusion

This study successfully implemented a complete semantic segmentation pipeline for weed detection in soybean fields using two different approaches.

While Roboflow provided a streamlined interface for annotation and baseline modeling, the Colab SegFormer approach demonstrated superior performance through deeper architectures and customized loss functions.

The best-performing model achieved $F1 = 0.8411$ and $IoU = 0.7672$, highlighting the potential of transformer-based models for agricultural weed mapping.